



Knowledge-Enhanced Multi-Behaviour Contrastive Learning for Effective Recommendation

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ABSTRACT

Real-world recommendation scenarios usually need to handle diverse user-item interaction behaviours, including page views, adding items into carts, and purchasing activities. The interactions that precede the actual target behaviour (e.g., purchasing an item) allow to capture the user's preferences from different angles, and are used as auxiliary information (e.g., page views) to enrich the system's knowledge about the users' preferences, thereby helping to enhance recommendation for the target behaviour. Despite efforts in modelling the users' multi-behaviour interaction information, the existing multi-behaviour recommenders still face two challenges: (1) Data sparsity across multiple user behaviours is a common issue that limits the recommendation performance, particularly for the target behaviour, which typically exhibits fewer interactions compared to other auxiliary behaviours; (2) Noisy auxiliary interactive behaviours where the information in the auxiliary behaviours might be non-relevant for recommendation. In this case, a direct adoption of contrastive learning between the target behaviour and the auxiliary behaviours will amplify the noise in the auxiliary behaviours, thereby negatively impacting the real semantics that can be derived from the target behaviour. To address these two challenges, we propose a new model called Knowledge-Enhanced Multi-behaviour Contrastive Learning for Recommendation (KEMCL). In particular, to address the problem of sparse user multi-behaviour interaction information, we leverage a dual-perspective knowledge encoding component that enriches the semantic representations of items, and generate supervision signals through self-supervised learning so as to enhance recommendation. In addition, we develop a cross-behaviour learning component, which includes two contrastive learning (CL) methods, inter CL and intra CL, to alleviate the problem of noisy auxiliary interactions. Extensive experiments on three public recommendation datasets show that our proposed KEMCL model significantly outperforms seven existing state-of-the-art methods. In particular, our KEMCL model outperforms the best baseline, namely KMCLR, by 5.42% on the large Tmall dataset.

CCS CONCEPTS

• **Information systems** → **Recommender systems.**



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KEYWORDS

Multi-behaviour Recommendation, Knowledge Graph, Contrastive Learning

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1 INTRODUCTION

Collaborative filtering (CF) is a commonly used paradigm for obtaining user preferences from historical interaction behaviours. However, most CF-based models [10, 19, 23, 38] treat all types of user interactions as a single type of behaviour. Hence, such models miss the opportunity to capture the fine-grained behaviours of users through various interaction types. Indeed, in typical recommendation scenarios, users often exhibit different types of behaviours with items. For example, on online shopping platforms, while interacting with the system, users might "add to cart", "mark as like", and "view" items, yet might not proceed with the final decision to "purchase". These different types of interactions serve as auxiliary behaviours, providing additional context to help the recommender system to learn the target behaviour (e.g., the purchase of an item). Early recommender systems [3, 5] modelled such *multi-behaviour* information by using neural collaborative filtering approaches. Later, MBGCN [11] performed multi-behaviour modelling on a user-item interaction graph consisting of graph neural networks (GNNs) in order to improve the recommendation performance. However, the aforementioned methods [3, 5, 11] directly use the available multi-behaviours information, and are hence limited by the problem of data sparsity across multiple user behaviours. Indeed, in most real-world recommendation scenarios, multi-behaviour interaction information remains sparse [30, 32], especially for the target behaviours, which typically exhibit fewer interactions compared to other auxiliary behaviours. Knowledge graphs (KG) [9, 15] and contrastive learning (CL) approaches [17, 24, 28] have previously been proposed to address the data sparsity problem. KG-based models [21, 25, 41] enrich the item representations by adding semantic information to candidate items. In addition, contrastive learning approaches [32, 34] generate supervision signals from both unlabelled multi-behaviour graphs and knowledge graphs, thereby allowing a deeper learning of the commonalities and differences between users and items on these two graphs. For example, KMCLR [32] used a knowledge graph to enrich item representations and leveraged contrastive learning to model the coarse- and fine-grained differences in users' behaviours for an improved performance. However, these

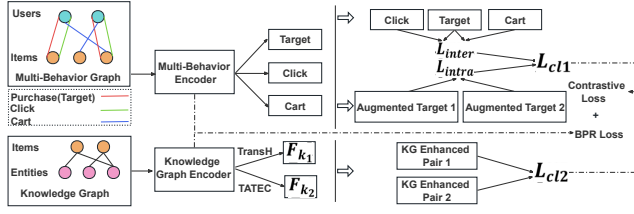


Figure 1: The architecture of our proposed KEMCL model.

KG and CL approaches are limited by the noise encapsulated in the auxiliary interactive behaviours [27]. Indeed, while the auxiliary behaviours provide useful information, they also contain noisy interactions, which complicate the modelling of the target behaviour. Consequently, we argue that directly applying contrastive learning approaches to multi-behaviour information can further amplify the noisy signal extracted from the auxiliary behaviours, thereby impeding the accurate learning of the target behaviour’s true semantics.

To address the data sparsity problem across multiple user behaviours and noisy auxiliary interactive behaviours, we propose a new recommendation model, called **Knowledge-Enhanced Multi-behaviour Contrastive Learning (KEMCL)**. Specifically, we propose a dual-perspective knowledge encoding component, which uses two knowledge-enhanced representation learning methods, namely (TransH and TATEC), to enrich the items’ representations, thus addressing the data sparsity problem across multiple user behaviours. Furthermore, we propose a cross-behaviour learning component that acts at two levels: introducing rich auxiliary signals in the auxiliary behaviours (inter CL), and generating additional intra supervision signals in the target behaviour (intra CL), thus addressing the problem of noisy auxiliary behaviours. To the best of our knowledge, we are the first to use a combination of dual-perspective KG-enhanced item representations and multi-behaviour generating supervision signals from two levels (i.e. inter CL and intra CL) to improve the recommendation performance based on multi-behaviour. To evaluate our proposed KEMCL model, we conduct extensive experiments on three public datasets. We show that KEMCL significantly outperforms seven existing state-of-the-art methods, in particular outperforming KMCLR by 5.42% on the large Tmall dataset.

2 RELATED WORK

We position our work in relation to two related fields, namely knowledge graph-based and multi-behaviour recommendation.

Knowledge Graph-based Recommendation: Knowledge graphs (KGs) have been widely used in enhancing user-item interaction behaviours due to their rich item semantic information [1, 9, 15, 40]. KG-based recommenders often use the translation distance method for knowledge-enhanced item representations, which learns low-dimensional embeddings of entities, items, and relations from the knowledge graph by minimising the transformation error. For example, KGAT [22] used a translation distance method called TransR [14] to obtain knowledge-enhanced item representations, and introduced a graph attention mechanism to enhance the user and item embeddings. Later, KGCL [34] also used a translation distance method to enhance the KG information, but integrated instead a contrastive learning method with a KG and a user-item

interaction graph to address the data sparsity problem. Furthermore, other semantic matching methods have been used to learn the potential semantics between entities and items in a KG. For example, TATEC [6] used a two-way (entity-relation, relation-item) and a three-way (entity-relation-item) combination to enhance the item representations from a knowledge graph. Different from the aforementioned recommender systems, which solely use a single knowledge-enhanced perspective, our KEMCL model uses both the translation distance and semantic matching methods to generate knowledge-enhanced item representations. This dual-method strategy enables a more effective contrastive learning, which contrasts knowledge-enhanced item representations to emphasise the useful KG information in the final user/item embeddings, thereby addressing the problem of data sparsity across multiple user behaviours.

Multi-behaviour Recommendation: Existing multi-behaviour recommenders [16, 43] mainly focus on leveraging auxiliary behaviours information captured from interactions between the users and items on order to estimate the users’ preferences from various perspectives, thereby refining the recommendation accuracy for the target behaviour. MATN [29] designed a transformer-based encoder to learn the direct dependencies among multiple behaviours. Later, researchers introduced graph neural networks (GNNs) to model multi-hop long-distance information transmission between the users and items [30, 31, 39]. For example, MBGCN [11] proposed a graph convolution network to learn the importance of auxiliary behaviours to the target behaviour as well as to extract behavioural semantics from the multi-behaviours data. Furthermore, several multi-behaviour recommenders [8, 33] applied contrastive learning (CL) to multi-behaviours interaction information in order to enhance users’ and items’ feature representations. However, these recommenders suffer from the noise encountered in the auxiliary behaviours. We argue that this issue necessitates the integration of additional supervision signals to enhance the recommendation accuracy [18, 27, 32, 37]. Therefore, we integrate a cross-behaviour learning component with two CL methods, inter CL and intra CL, in our KEMCL model. Inter CL aims to use the richer supervision signals generated from the auxiliary behaviours for an effective representation of the target behaviour, while intra CL aims to produce additional supervision signals in the target behaviour. We leverage these two CL methods to explicitly contrast the positive and negative interactions, enabling our KEMCL model to identify and emphasise essential interactions, so as to mitigate the negative effect of noise in the auxiliary behaviours.

3 METHODOLOGY

In this section, we first introduce the used notations and the recommendation task (Section 3.1). Figure 1 illustrates the general architecture of our proposed KEMCL model. In Sections 3.2 and 3.3, we describe the proposed cross-behaviour learning and the dual-perspective knowledge encoding components of our KEMCL model. Then, we introduce the used loss functions in our KEMCL model (Section 3.4). Finally, we analyse the computational complexity of our model in Section 3.5.

3.1 Task Formulation

We define $\mathcal{U}$ and $\mathcal{I}$ to represent the users set and the items set, respectively: $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$ and $\mathcal{I} = \{i_1, i_2, \dots, i_N\}$. We use

a knowledge graph based on the item information, and construct a multi-behaviour graph based on the user-item interactions. We define the multi-behaviour graph as $\mathcal{G}_u = \{\mathcal{U}, \mathcal{Y}, \mathcal{I}\}$, where $\mathcal{Y} = \{y^1, \dots, y^K\}$ represents the edges set of K behaviours, and $\mathcal{V} = \mathcal{U} \cup \mathcal{I}$ represents the nodes set. Specifically, if $y_{u,i}^k = 1$, the user and item have a k -type interaction behaviour; otherwise, $y_{u,i}^k = 0$ indicates the absence of such an interaction. In order to leverage the rich item-related information in the knowledge graph, we construct the knowledge graph as $\mathcal{G}_k = \{(h, r, t)\}$, where each triplet represents a direct relation r between the head entity h and the tail entity t . The studied task is defined as follows: **Input**: a multi-behaviour graph and a knowledge graph. **Output**: a prediction function that estimates the probability $y_{u,i}^k$ of a user interacting with an item given a k -type behaviour.

3.2 Cross-behaviour Learning

Existing multi-behaviour recommenders mainly focus on capturing multi-hop information between nodes within the user-item interactions graph, which encompasses all types of behaviours [32]. Using a user multi-behaviour interaction graph as input, we employ a multi-behaviour encoder (shown top-middle in Figure 1) based on LightGCN [10] to learn user/item representations of a single type of behaviour:

$$x_u^{k,(d+1)} = \sum_{i \in N_u^k} x_i^{k,d}, \quad x_i^{k,(d+1)} = \sum_{u \in N_i^k} x_u^{k,d}, \quad (1)$$

where $x_u^{k,d}$ and $x_i^{k,d}$ are the user and item representations obtained under behaviour k at layer d . N_u^k and N_i^k represent the neighbour nodes of the user and item, respectively. As a result, we obtain the user/item representations from each single type of behaviour. In order to capture the dependencies between different behaviours, we use a multi-layer transformation matrix to model the cross-behaviours of the user representations:

$$x_u = \text{PReLU}(W_1 \cdot (x_u^0, \dots, \|x_x^d\|, \dots, \|x_u^D\|)), \quad (2)$$

$$x_u^d = \sigma(W_2 \cdot \frac{\sum_{k \in K} x_u^{k,(d+1)}}{K}), \quad (3)$$

where $W^1 \in \mathbb{R}^{D \times l}$ and $W^2 \in \mathbb{R}^{l \times l}$ are the weighted matrices for the behaviours, l is the output dimension and PReLU is the activation function. Similarly, we obtain the item representations. As such, we obtain the user/item representations that capture the personalised behavioural correlations via cross-behaviour modelling.

As mentioned in Section 1, recommenders can make use of the auxiliary behaviours to derive additional information to better learn the target behaviour. Moreover, the introduced auxiliary behaviours have noisy interactions, which need adequate processing to alleviate the noise within these behaviours. Therefore, we develop a cross-behaviour learning component, which performs CL at both the inter-behaviour and intra-behaviour levels, to enrich the target behaviour and mitigate the negative effect of noisy auxiliary behaviours. For inter CL, since there are more supervision signals in the auxiliary behaviours, we perform contrastive learning between the target behaviour and other auxiliary behaviours to enable knowledge transfer from the auxiliary behaviours to the target behaviour. For all auxiliary behaviours k from $\{1, 2, \dots, K-1\}$, we conduct contrastive learning between the target behaviour K and

the auxiliary behaviour k . We select identical users/items from two behaviour subgraphs as positive pairs (i.e., $\{(\mathbf{x}_{n,K}, \mathbf{x}_{n,k}) | n \in \mathcal{V}\}$), and different users/items from two behaviour subgraphs as negative pairs (i.e., $\{(\mathbf{x}_{n,K}, \mathbf{x}_{n',k}) | n \in \mathcal{V}\}$). We design a multi-behaviour inter contrastive loss based on InfoNCE [2]:

$$\mathcal{L}_{inter} = \sum_{k=1}^K \sum_{n \in \mathcal{V}} -\log \frac{\exp(\phi(\mathbf{x}_{n,K}, \mathbf{x}_{n,k})/\tau)}{\sum_{n' \in \mathcal{V}_{n' \neq n}} \exp(\phi(\mathbf{x}_{n,K}, \mathbf{x}_{n',k})/\tau)}, \quad (4)$$

where τ represents the temperature parameter in the softmax function. $\phi(\cdot)$ denotes the inner products to determine the similarity between the positive and negative pairs. Consequently, our KEMCL model produces additional supervision signals from the auxiliary behaviours, thus enriching the representation of the target behaviour.

On the other hand, as discussed in Section 2, the auxiliary behaviours contain noisy interactions, necessitating measures to prevent this noisy information from impacting the modelling of the target behaviour. Therefore, we develop an intra CL, which amplifies the impact of supervision signals within the target behaviour, thus preventing further noisy signals in the auxiliary behaviours being ingested in the cross-behaviour learning. Specifically, we first use dropout augmentation to generate two augmented subgraphs $\mathcal{G}_K^1$ and $\mathcal{G}_K^2$ from the target behaviour graph $\mathcal{G}_K = (\mathcal{V}_K, \mathcal{I}_K)$:

$$\mathcal{G}_K^1 = (\mathcal{V}_K, \mathbf{M}_1 \odot \mathcal{I}_K), \quad \mathcal{G}_K^2 = (\mathcal{V}_K, \mathbf{M}_2 \odot \mathcal{I}_K), \quad (5)$$

where $\mathbf{M}_1$ and $\mathbf{M}_2$ are mask vectors. We then construct intra CL by contrasting the users'/items' representations in $\mathcal{G}_K^1$ and $\mathcal{G}_K^2$, and calculate the loss $\mathcal{L}_{intra}$ similar to Equation (4). Then, we combine $\mathcal{L}_{inter}$ and $\mathcal{L}_{intra}$ together to obtain the final CL loss of cross-behaviour learning as $\mathcal{L}_{cl1} = \mathcal{L}_{inter} + \mathcal{L}_{intra}$.

3.3 Dual-perspective Knowledge Encoding

Knowledge graphs contain item-related information that could enrich the item's semantic information for recommendation [4, 9, 15, 36]. We develop a knowledge graph encoder (shown in the middle of the bottom row of Figure 1) to obtain the item representation from the KG:

$$\mathbf{z}_i^{(d+1)} = \frac{\sum_{(r,e) \in \mathcal{N}_i} \mu(i, r, e) \mathbf{z}_r \odot \mathbf{z}_e^{(d)}}{|\mathcal{N}_i|}, \quad (6)$$

$$\mu(i, r, e) = \frac{\exp((\mathbf{z}_i \| \mathbf{z}_r)^T \cdot (\mathbf{z}_e \| \mathbf{z}_r))}{\sum_{(e',r) \in \hat{\mathcal{N}}(i)} \exp((\mathbf{z}_i \| \mathbf{z}_r)^T \cdot (\mathbf{z}_{e'} \| \mathbf{z}_r))}, \quad (7)$$

where $\mathbf{z}_i$ represents the item representation, (i, r, e) represents the entity-relation-item triplet in the knowledge graph, d represents the number of layers, and $\hat{\mathcal{N}}(i)$ represents the neighbouring entities $\mathcal{N}_i$. Similar to Section 3.2, we aim to generate two knowledge-enhanced contrastive pairs on the KG and two target behaviour subgraphs to mine rich item semantics. In this work, we propose to use two embedding methods to obtain the KG representations, namely the translation distance method TransH [26] and the semantic matching method TATEC [6], thereby providing two different perspectives on the semantics of items. These two methods focus on the relational and latent semantics of the triplets in the KG, thereby obtaining the triplet scores with different perspectives on the semantics of items:

$$\alpha(i, r, e) = -\|(\mathbf{z}_i - W_r^T \mathbf{z}_i W_r) + d_r - (\mathbf{z}_e - W_r^T \mathbf{z}_e W_r)\|_2^2, \quad (8)$$

Table 1: Statistics of datasets.

Dataset	Users	Items	Interaction	Density	Interactive behaviour Type
Tmall	31,882	31,232	1,451,219	0.145%	{Page-View, Favourite, Cart, Purchase}
IJCAI	22,438	35,573	1,779,195	0.223%	{Click, Cart, Favourite, Purchase}
Retail	2,174	30,113	97,381	0.149%	{Page-View, Cart, Purchase}

$$\beta(i, r, e) = z_i^T M_r z_e + z_i^T z_r + z_e^T z_r + z_i^T d D z_e. \quad (9)$$

where d_r is the relation vector between z_i and z_e in the hyperplane, W_r is the norm vector of the hyperplane, M_r is the projection matrix, and D is a multi-behaviour diagonal matrix. Using these two methods, we obtain the scores of the KG triplets from different perspectives, and employ these scores to construct two graphs ($\mathcal{F}_{k1}$ and $\mathcal{F}_{k2}$) by using the scores as edge weights.

While a knowledge graph introduces useful information to enhance the recommendation performance, it might also add non-relevant information to the item representations [20]. Since [34] observed that a higher structural consistency score of items on the KGs leads to an effective recommendation performance, we calculate another score on $\mathcal{F}_{k1}$ to guide the generation of an enhanced target behaviour subgraph:

$$c^1 = s(f_1(z_i), f'_1(z_i)), \quad (10)$$

where f_1 and f'_1 are the sampled subgraphs obtained by setting different seeds in $\mathcal{F}_{k1}$, and $s(\cdot)$ is the similarity function. Similarly, we obtain the structural consistency score c^2 of $\mathcal{F}_{k2}$. After obtaining the item consistency scores c^1 and c^2 , we use the resulting probabilities to construct enhanced target behaviour subgraphs:

$$p_1 = p_a \cdot \mu_{p'} \cdot \max\left(\frac{\exp(c^1) - \min(w_{i,u})}{\max(w_{i,u}) - \min(w_{i,u})}, p_\zeta\right), \quad (11)$$

where p_1 represents the calculated dropout probability of an user-item interaction related to $\mathcal{F}_{k1}$, p_a control the strength based on the average value, $\mu_{p'}$ is the mean value, $w_{i,u}$ represents the influence degree of item i on user u in subgraph $\mathcal{F}_{k1}$ and p_ζ is the truncation probability. The calculated value of $w_{i,u}$ is proportional to the structural consistency score c^1 . Then, we perform a min-max normalisation on $\exp(c^1)$, which means normalising $\exp(c^1)$ to the range defined by $\min(w_{i,u})$ and $\max(w_{i,u})$, where $\min(w_{i,u})$ and $\max(w_{i,u})$ are the minimum and maximum values of $w_{i,u}$, respectively, across all users. Similarly we obtain a p_2 related to $\mathcal{F}_{k2}$. We perform dropout augmentations to enhance the target behaviour subgraph respectively according to the two resulting probabilities, and obtain two enhanced target behaviour subgraphs $\mathcal{F}_{u1}$ and $\mathcal{F}_{u2}$. Through the above steps, we obtain two pairs of perspective graphs ($\mathcal{F}_{u1}, \mathcal{F}_{k1}$) and ($\mathcal{F}_{u2}, \mathcal{F}_{k2}$). Then, we further use contrastive learning between these two pairs to learn the users/items representations from KG and the target behaviour. We construct a knowledge-enhanced contrastive loss $\mathcal{L}_{cl2}$ similar to inter CL (Equation (4)). As a result, we introduce KG to enrich item representations and further apply CL on the KG and target behaviour graph to alleviate the problem of data sparsity.

3.4 Joint Learning

Once we obtain the user representations (x_u^{mul}, z_u^{kg}) and the item representations (x_i^{mul}, z_i^{kg}) through the proposed cross-behaviour

learning and dual-perspective knowledge encoding components, we set the parameter δ to combine these two components to obtain the final user representations $e_u = \delta z_u^{kg} + (1 - \delta)x_u^{mul}$ and item representations $e_i = \delta z_i^{kg} + (1 - \delta)x_i^{mul}$. We combine the two CL tasks and the recommendation task through a joint training strategy to optimise our KEMCL model. Our KEMCL model uses a combination of a cross-behaviour CL loss, a knowledge-enhanced CL loss and a Bayesian Personalised Ranking (BPR) loss for joint training:

$$\mathcal{L} = \sum_{u \in \mathcal{U}} \sum_{i \in N_u} \sum_{i' \notin N_u} -\log \sigma(e_u^T e_i - e_u^T e_{i'}) + \lambda_1 \mathcal{L}_{cl1} + \lambda_2 \mathcal{L}_{cl2} + \lambda_3 \|\Theta\|_2^2, \quad (12)$$

where N_u is the number of observed user interactions, and Θ denotes the learnable model parameters. λ_1 and λ_2 control the weights of the two contrastive learning losses, respectively.

3.5 Complexity analysis

We analyse the computational complexity of the two key components of our KEMCL model. The complexity of the dual-perspective knowledge encoding component is $O((|N_i| \cdot q) + v) \cdot l$ where $|N_i|$ is the number of neighbouring entities of item i , q is the number of nodes in the knowledge graph (KG), l is the hidden dimensionality, and v is the number of triplets. For the cross-behaviour learning component, its computational complexity is $O((|B| + M + N) \cdot K \cdot l \cdot d)$, where $|B|$ is the number of interactions, K is the types of behaviour, l is the hidden dimensionality, d is the network layers, M and N are the number of users and items respectively. Based on the above analysis, both our model and the best baseline (KMCLR [32]) have similar complexities, dominated by the KG-related encoding complexity.

4 EXPERIMENTS

We conduct experiments on three datasets to evaluate the performance of our proposed KEMCL model. We compare with seven existing state-of-the-art models and answer the following two questions: **RQ1**: How does KEMCL perform compared to other existing recommendation models? **RQ2**: How do the proposed components of KEMCL, namely the dual-perspective knowledge encoding and cross-behaviour learning, affect the recommendation performance?

Datasets and Evaluation Protocol: We conduct experiments on three public datasets to demonstrate the effectiveness of KEMCL (c.f. Table 1: Tmall [44], IJCAI [35] and Retail [13]). "Purchase" is the target behaviour in each dataset. We follow the same setup as [42] to build the knowledge graph during the experiments. We construct the training, validation, and test sets according to the ratio of 7:1:2. We use two representative metrics, Hit@10 and NDCG@10, to evaluate the effectiveness of our KEMCL model [27]. We use Xavier [7] and Normal [32] for the embeddings initialisation. We follow the same setting as [32], where the embedding size for all models is fixed to 32. We optimise KEMCL and the baselines during the training phase using the Adam [12] optimiser with a learning rate of 0.001. For the combination parameter δ , we vary it within {0.1, 0.2, 0.3, 0.4}. We adopt an L₂ regularisation, with the weight decay varied in {0.0005, 0.001, 0.005, 0.01}.

Baselines: We compare KEMCL with three groups of baselines: (1) collaborative filtering models (BPR [19], LightGCN [10]), (2) KG-based models (KGAT [22], KGCL [34]) and (3) multi-behaviour recommendation models (KHGT [30], CML [27], KMCLR [32]).

Performance Comparison (RQ1): Table 2 shows the performance comparison of our proposed KEMCL model with all used baselines. The best and the second-best performance are bolded and underlined, respectively. We use the Holm-Bonferroni corrected paired t-tests to show that all improvements over the baselines are statistically significant on the three used datasets. From Table 2, we observe that KEMCL significantly outperforms all baselines on both metrics on the three datasets. These improvements show the effectiveness of using the cross-behaviour learning and dual-perspective knowledge encoding components in our KEMCL model. Table 2 also shows that the multi-behaviour models (KHGT, CML and KMCLR) and KG-based models (KGAT, KGCL) are generally more effective than the collaborative filtering models (BPR, LightGCN) on the used datasets. This result indicates that it is beneficial to introduce additional knowledge (i.e., KG) and auxiliary behaviours (e.g., page views) to enrich the user/item representations. Furthermore, KMCLR, a recent recommendation model, which uses a KG and CL in multi-behaviour learning to obtain additional information, performs the best among the baselines although it is significantly outperformed by our KEMCL model. These results overall support the effectiveness of using KG and CL to address the data sparsity problem. Our KEMCL model further enriches the item representations from two perspectives, and uses inter and intra CLs to learn additional supervision signals for multi-behaviour. In answer to RQ1, we conclude that our KEMCL model successfully leverages the cross-behaviour learning and dual-perspective knowledge encoding components and significantly outperforms all strong baselines.

Ablation Study (RQ2): In order to study the impact of KEMCL's main components on the recommendation performance, we evaluate the following three variants of KEMCL: "w/o kg" is a variant that does not include the use of a KG in the items representations. "w/o inter" is a variant that removes the inter CL part of the cross-behaviour learning, while "w/o intra" removes the intra CL part of the cross-behaviour learning. Table 3 shows the results of the conducted ablation experiments. From Table 3, we observe that our KEMCL model clearly exhibits the best NDCG@10 and HR@10 performance across the three datasets in comparison to all its variants. The significant performance degradation observed in the "w/o inter" and "w/o intra" variants, which remove either inter CL or intra CL, shows the effectiveness of using supplementary supervision signals generated from the auxiliary behaviours and the benefits of internal supervision signals within the target behaviours, respectively. These results show that the application of inter CL and intra CL is successful on identifying useful supervision signals from the noisy behaviours, which in turn improves the recommendation performance. Table 3 also shows that removing the dual-perspective knowledge encoding component also leads to a significant performance degradation. This shows the effectiveness of KG in enhancing the item representations, allowing KEMCL to effectively address the problem of sparse behaviour data. In answer to RQ2, the results highlight the positive impacts of both the cross-behaviour learning and the dual-perspective knowledge encoding components within our KEMCL model on the recommendation performance.

Furthermore, as additional experiments, we conduct a sensitivity analysis of the cross-behaviour CL loss parameter, the knowledge-enhanced CL loss parameter and the temperature parameter, which

Table 2: Performance comparison results between KEMCL and the baselines on three datasets. * indicates that the results are significantly different from the KEMCL results using the Holm-Bonferroni corrected paired t-test with p-value<0.05.

Dataset	Tmall		IJCAI		Retail	
	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10
Methods						
BPR	0.2368*	0.1484*	0.1965*	0.1303*	0.2271*	0.1407*
LightGCN	0.3372*	0.2039*	0.2956*	0.1801*	0.2584*	0.1512*
KGAT	0.3791*	0.2207*	0.3346*	0.2014*	0.2983*	0.1804*
KGCL	0.3955*	0.2284*	0.3514*	0.2128*	0.3092*	0.1820*
KHGT	0.3967*	0.2320*	0.3523*	0.2185*	0.3186*	0.1903*
CML	0.5119*	0.3014*	0.4851*	0.2811*	0.3374*	0.2042*
KMCLR	0.5614*	0.3432*	0.5301*	0.3212*	0.3461*	0.2186*
KEMCL	0.5918	0.3637	0.5519	0.3353	0.3576	0.2251

Table 3: Results of the ablation study. The * indicates that the result is significantly different from KEMCL using the Holm-Bonferroni corrected paired t-test with p-value<0.05.

Dataset	Tmall		IJCAI		Retail	
	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10
Methods						
w/o - inter	0.5782*	0.3545*	0.5407*	0.3296*	0.3471*	0.2191*
w/o - intra	0.5841*	0.3585*	0.5476*	0.3328*	0.3541*	0.2226*
w/o - kg	0.5844*	0.3586*	0.5480*	0.3329*	0.3549*	0.2232*
KEMCL	0.5918	0.3637	0.5519	0.3353	0.3576	0.2251

we now briefly summarise. For the cross-behaviour CL loss parameter λ_1 and the knowledge-enhanced CL loss parameter λ_2 , we vary from 0.01 to 0.10 with a step of 0.01. The performance of KEMCL is overall stable, with the best performance obtained with λ values around 0.04 - 0.06, but effectiveness is reduced for particularly high and low values of λ_1 and λ_2 . This suggests that the setting of the two loss parameter needs to be carefully chosen to match the corresponding loss recommendation task to improve the performance of the model. For the temperature parameter τ , we vary from 0.1 to 0.5 with a step of 0.1. KEMCL performs the best with a value of 0.2 on the three used datasets. The performance drops for particularly higher values of τ . This result suggests that when $\tau = 0.2$, the model can effectively distinguish between positive and negative samples to achieve the best performance.

5 CONCLUSIONS

We proposed a new recommendation model, called KEMCL, to address both the problem of data sparsity across multiple user behaviours and the noisy auxiliary interactive behaviour. Specifically, we used a dual-perspective knowledge encoding component to address the data sparsity problem across multiple user behaviours and leveraged a cross-behaviour learning component to alleviate the noise encountered within the auxiliary interactive behaviour problem. Our extensive experiments using three datasets and seven strong baselines showed the effectiveness of our KEMCL model. In particular, KEMCL significantly outperformed the strongest baseline, namely KMCLR, by up to 5.42%. In addition, our ablation experiments demonstrated the positive impact of both the dual-perspective knowledge encoding and the cross-behaviour learning components on the recommendation performance of KEMCL.

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